ATAR • YEAR 12 CHEMISTRY

REDOX & Electrochemistry

Complete Study Notes — Unit 3

Oxidation & Reduction

Oxidation Numbers

Half-Equations

Galvanic Cells

Standard Potentials

Electrolysis

Faraday Calculations

Corrosion

Everything you need for the exam — definitions, worked examples, data sheet & exam traps.

Contents

1 What is Redox?

A **redox reaction** is a reaction in which electrons are transferred from one species to another. Every redox reaction is two processes happening at the same time: **oxidation** and **reduction**. One cannot occur without the other.

KEY DEFINITIONS (LEARN THE ELECTRON DEFINITIONS)

- **Oxidation** = *loss* of electrons (oxidation number increases).
- **Reduction** = *gain* of electrons (oxidation number decreases).

OIL RIG

Oxidation **I**s **L**oss · **R**eduction **I**s **G**ain (of electrons)

Three ways to describe redox

Process	Electrons	Oxygen	Hydrogen	Oxidation number
Oxidation	lost	gained	lost	increases (more +)
Reduction	gained	lost	gained	decreases (more -)

The electron-transfer definition is the most general and the one you should always use at ATAR level. The oxygen/hydrogen definitions are useful shortcuts (e.g. in organic and combustion reactions).

THE GOLDEN RULE

Oxidation and reduction always occur **together** — the electrons lost by the species being oxidised are exactly the electrons gained by the species being reduced. Electrons are never created or destroyed.

2 Oxidation Numbers (Oxidation States)

The **oxidation number** is a signed number assigned to an atom that represents the charge it *would* have if all bonds were 100% ionic. It is the bookkeeping tool used to track electron transfer and identify what is oxidised and reduced.

RULES FOR ASSIGNING OXIDATION NUMBERS (APPLY IN ORDER)

1. An atom in its **free element** has an oxidation number of **0** (e.g. Na, O_2, Cl_2, S_8, P_4).
2. For a **monatomic ion**, the oxidation number equals the ionic charge (e.g. Na^+ = +1, S^{2-} = -2, Al^{3+} = +3).
3. **Group 1** metals = +1; **Group 2** metals = +2; **Aluminium** = +3 (always in compounds).
4. **Fluorine** = -1 in all its compounds.
5. **Hydrogen** = +1 (except -1 in metal hydrides such as NaH, CaH_2).
6. **Oxygen** = -2 (except -1 in peroxides e.g. H_2O_2; +2 in OF_2; -½ in superoxides).
7. The sum of oxidation numbers = **0** for a neutral compound, or = the **overall charge** for a polyatomic ion.

Worked examples — find the unknown oxidation number

EXAMPLE — S IN SULFATE, SO_4^{2-}

Let S = *x*. Oxygen = -2 (×4). Sum must equal the ion charge (-2):

$x + 4(-2) = -2 \qquad x - 8 = -2 \qquad \mathbf{x = +6}$

EXAMPLE — MN IN PERMANGANATE, MNO_4^-

$x + 4(-2) = -1 \qquad x = +7$ (**Mn = +7**)

EXAMPLE — CR IN DICHROMATE, $CR_2O_7^{2-}$

$2x + 7(-2) = -2 \qquad 2x = +12 \qquad$ **Cr = +6**

USING OXIDATION NUMBERS TO IDENTIFY REDOX

Compare oxidation numbers of each element **before and after** the reaction:

- If an oxidation number **increases** that element was **oxidised**.
- If an oxidation number **decreases** that element was **reduced**.
- If **no** oxidation number changes the reaction is **not** redox (e.g. most acid–base and precipitation reactions).

Disproportionation

A special redox reaction in which the **same element** is *simultaneously* oxidised and reduced. Example — chlorine in water:

$Cl_2 + H_2O \quad HOCl + HCl$ (Cl goes 0 +1 and 0 -1)

3 Oxidising & Reducing Agents · Conjugate Pairs

AGENTS

- An **oxidising agent (oxidant)** *causes* oxidation by *accepting* electrons — so it is itself **reduced**.
- A **reducing agent (reductant)** *causes* reduction by *donating* electrons — so it is itself **oxidised**.

DON'T GET CAUGHT OUT

The oxidising agent is the species that is **reduced**; the reducing agent is the species that is **oxidised**. Students frequently swap these — read the sentence twice.

Conjugate redox pairs

Just as acids have conjugate bases, every oxidant/reductant forms a **conjugate redox pair** linked by electron transfer:

oxidised form + n e^- reduced form

e.g. Cu^{2+} / Cu and Zn^{2+} / Zn. The stronger the oxidant, the weaker its conjugate reductant (and vice versa).

Common oxidants and reductants seen at ATAR

Strong oxidising agents	Strong reducing agents
MnO_4^- (acidified), $Cr_2O_7^{2-}$, Cl_2, F_2, H_2O_2, concentrated HNO_3	Reactive metals (Li, K, Na, Mg, Al, Zn), H_2, C, SO_2, I^-

4 Writing & Balancing Half-Equations

A **half-equation** shows either the oxidation *or* the reduction process on its own, including the electrons transferred. Electrons appear on the **left** of a reduction half-equation and on the **right** of an oxidation half-equation.

BALANCING A HALF-EQUATION (ACIDIC SOLUTION) — 5 STEPS

1. Balance all atoms **except O and H**.
2. Balance **O** by adding **H_2O**.
3. Balance **H** by adding **H^+**.
4. Balance **charge** by adding **electrons (e^-)**.
5. Check atoms *and* charge both balance.

WORKED EXAMPLE — PERMANGANATE REDUCED IN ACID

Half-reaction: MnO_4^- Mn^{2+} 1. Mn balanced.

2. Balance O with water: MnO_4^- $Mn^{2+} + 4H_2O$

3. Balance H with H^+: $MnO_4^- + 8H^+$ $Mn^{2+} + 4H_2O$

4. Balance charge (left = +7, right = +2)   add $5e^-$ to left:

$MnO_4^- + 8H^+ + 5e^-$ $Mn^{2+} + 4H_2O$

BALANCING IN BASIC (ALKALINE) SOLUTION

Balance as if acidic first, then **add OH^- to both sides** equal to the number of H^+. Combine $H^+ + OH^-$ H_2O and cancel any water that appears on both sides.

5 Combining Half-Equations into a Full Equation

METHOD

1. Write the **oxidation** half-equation and the **reduction** half-equation.
2. Multiply each by a whole number so the **electrons are equal** in both.
3. Add the two half-equations together and **cancel electrons** (and any species appearing identically on both sides, e.g. H^+, H_2O).

WORKED EXAMPLE — MNO_4^- OXIDISING FE^{2+} IN ACID

Reduction (×1): $MnO_4^- + 8H^+ + 5e^-$ $Mn^{2+} + 4H_2O$

Oxidation (×5): $5Fe^{2+}$ $5Fe^{3+} + 5e^-$

Overall: $MnO_4^- + 8H^+ + 5Fe^{2+}$ $Mn^{2+} + 5Fe^{3+} + 4H_2O$ e^- cancel (5 each)

Check: charge on left = (-1)+(+8)+(+10) = +17; right = (+2)+(+15) = +17

6 Galvanic (Voltaic) Cells

A **galvanic cell** converts chemical energy into electrical energy using a **spontaneous** redox reaction. The oxidation and reduction reactions are physically separated into two **half-cells**, forcing the electrons to travel through an external wire — this electron flow is the electric current.

Components & their roles

Component	Role
Anode	Electrode where **oxidation** occurs. In a galvanic cell it is the **negative** terminal.
Cathode	Electrode where **reduction** occurs. In a galvanic cell it is the **positive** terminal.
Salt bridge	Contains an inert electrolyte (e.g. KNO_3). Completes the circuit and maintains **electrical neutrality** by allowing ion migration between half-cells.
External circuit (wire)	Carries electrons from anode cathode. May include a voltmeter or load.
Electrolyte solutions	Contain the ions of each half-cell that are involved in the electrode reactions.

MEMORY AIDS

- **AN OX** — **AN**ode = **OX**idation.
- **RED CAT** — **RED**uction = **CAT**hode.
- Electrons flow **A** **C** (Anode to Cathode) through the wire, always.
- Polarity: in a **galvanic** cell anode = **-**, cathode = **+** (this is reversed in an electrolytic cell).

Direction of movement

- **Electrons:** anode external wire cathode.
- **Conventional current:** opposite to electrons (cathode anode in the wire).
- **Anions** (-) in the salt bridge migrate **toward the anode**; **cations** (+) migrate **toward the cathode** — to balance the charge building up as ions are produced/consumed.

CLASSIC EXAMPLE — THE DANIELL CELL (ZN | CU)

Anode (oxidation): $Zn(s) \quad Zn^{2+}(aq) + 2e^-$ $E° = -0.76$ V

Cathode (reduction): $Cu^{2+}(aq) + 2e^- \quad Cu(s)$ $E° = +0.34$ V

Overall: $Zn(s) + Cu^{2+}(aq) \quad Zn^{2+}(aq) + Cu(s)$ $E°_{cell} = +1.10$ V

The Zn electrode loses mass (dissolves); the Cu electrode gains mass; $[Cu^{2+}]$ falls and $[Zn^{2+}]$ rises.

Cell diagram notation (shorthand)

Convention: **anode on the left**, cathode on the right. A single line **|** = phase boundary; a double line **||** = salt bridge.

Zn(s) | Zn^{2+}(aq) || Cu^{2+}(aq) | Cu(s)

If a half-cell has no solid conductor (e.g. Fe^{3+}/Fe^{2+}), an **inert electrode** such as platinum or graphite is used, e.g. Pt | Fe^{2+}, Fe^{3+}.

7 Standard Electrode Potentials & the Electrochemical Series

Each half-cell has a tendency to gain electrons, measured as its **standard reduction potential (E°)** in volts. Potentials are measured relative to the **Standard Hydrogen Electrode (SHE)**, which is defined as exactly **0.00 V**.

STANDARD CONDITIONS (THE "°" SYMBOL)

- Concentration of all solutions = **1.0 mol L^{-1}**
- Pressure of all gases = **100 kPa (1 bar)**
- Temperature = **25 °C (298 K)**
- Measured against the SHE (Pt electrode, H_2 gas, 1 mol L^{-1} H^+).

READING THE TABLE (ALL VALUES WRITTEN AS REDUCTIONS)

- A **more positive** E° the species is a **stronger oxidising agent** (gains electrons readily; e.g. F_2, MnO_4^-).
- A **more negative** E° the reduced form is a **stronger reducing agent** (loses electrons readily; e.g. Li, K, Na).
- The **strongest oxidant** sits **top-left**; the **strongest reductant** sits **bottom-right** of the table.

THE "ANTICLOCKWISE / CLOCKWISE" PREDICTION RULE

Write the two relevant half-equations with the more positive E° on top. The reaction that proceeds is: **top half-equation goes forward (reduction)** and **bottom half-equation goes in reverse (oxidation)**. Trace an anticlockwise loop: the oxidant (top-left) reacts with the reductant (bottom-right).

8 Calculating Cell EMF & Predicting Reactions

STANDARD CELL POTENTIAL (EMF)

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

Both values are taken directly as **reduction potentials** from the table — do **not** change the sign of the anode value yourself; the minus sign in the formula does that. (Equivalently: $E^\circ_{cell} = E^\circ_{reduction} + E^\circ_{oxidation}$, where the oxidation value has its sign reversed.)

WHAT THE SIGN OF $E°_{CELL}$ TELLS YOU

- **$E°_{cell} > 0$** reaction is **spontaneous** as written (works as a galvanic cell). Corresponds to $\Delta G < 0$.
- **$E°_{cell} < 0$** **non-spontaneous** (would need an external power supply — electrolysis).

Link to free energy: $\Delta G° = -nFE°_{cell}$, where n = mol of electrons and F = 96 500 C mol^{-1}.

WORKED EXAMPLE — WILL SILVER REACT WITH COPPER METAL?

Half-cells: $Ag^{+} + e^{-}$ Ag (E° = +0.80 V) and $Cu^{2+} + 2e^{-}$ Cu (E° = +0.34 V).

The higher E° (Ag^{+}) is reduced cathode; Cu is oxidised anode.

$E°_{cell} = E°_{cathode} - E°_{anode}$ = (+0.80) - (+0.34) = **+0.46 V** (positive spontaneous) Overall: Cu(s) + $2Ag^{+}(aq)$ $Cu^{2+}(aq)$ + 2Ag(s).

E° DOES NOT DEPEND ON HOW MUCH YOU HAVE

E° is an **intensive** property. If you multiply a half-equation by a coefficient to balance electrons, you do **not** multiply its E° value. Voltage does not scale with amount of substance.

Limitations of predictions from E° values

- Predictions assume **standard conditions**; changing concentration, temperature or pressure shifts the actual potential.
- E° predicts whether a reaction is *thermodynamically* favourable, **not the rate** — a spontaneous reaction may be immeasurably slow (kinetics).
- **Overpotential** can mean the predicted product is not the one observed (important in electrolysis, §10).

9 Electrolytic Cells & Electrolysis

Electrolysis uses electrical energy from an external power source to drive a **non-spontaneous** redox reaction ($E°_{cell} < 0$). It is the reverse energy conversion of a galvanic cell: electrical energy   chemical energy.

Key differences from a galvanic cell

Feature	Galvanic cell	Electrolytic cell
Energy change	Chemical electrical	Electrical chemical
Spontaneity	Spontaneous ($E°_{cell} > 0$)	Non-spontaneous (needs power supply)
Half-cells	Two separate half-cells + salt bridge	Usually one container, two electrodes in the same electrolyte
Anode polarity	Negative (-)	**Positive (+)**
Cathode polarity	Positive (+)	**Negative (-)**

WHAT STAYS THE SAME IN BOTH CELL TYPES

- Oxidation **always** occurs at the **anode**; reduction **always** at the **cathode**.
- Electrons **always** travel from anode   cathode in the external circuit.

It is the **polarity (+/-) that flips** between the two cell types, not the location of oxidation/reduction.

Electrolysis of a molten salt (simplest case)

Molten ionic compounds contain only the compound's ions — no water to compete.

MOLTEN NACL (DOWNS PROCESS)

Cathode (-): $Na^+ + e^-$ $Na(l)$

Anode (+): $2Cl^-$ $Cl_2(g) + 2e^-$

Products: sodium metal at the cathode, chlorine gas at the anode.

10 Predicting the Products of Electrolysis

In **aqueous** solutions, **water can be oxidised or reduced** in competition with the dissolved ions. You must compare the possible reactions at each electrode.

WATER'S COMPETING HALF-REACTIONS

Reduction (at cathode): $2H_2O + 2e^-$ $H_2(g) + 2OH^-$ $E° = -0.83$ V

Oxidation (at anode): $2H_2O$ $O_2(g) + 4H^+ + 4e^-$ $E° = -1.23$ V (as an oxidation)

HOW TO DECIDE THE PRODUCT AT EACH ELECTRODE

1. List every species that could react at that electrode (ions + water).
2. **Cathode (reduction):** the species with the **highest (most positive) E°** is reduced most easily.
3. **Anode (oxidation):** the species that is oxidised most easily is the one whose reduction E° is **most negative** (i.e. best reducing agent).
4. Then apply the practical factors below — the E° prediction is a starting point only.

Three factors that affect the actual product

Factor	Effect
Standard potential (E°)	The primary predictor — most easily reduced/oxidised species reacts.
Concentration	A high concentration of an ion can cause it to be discharged in preference to water (e.g. concentrated Cl^- gives Cl_2, not O_2). Predicted by Le Châtelier / Nernst shifts.
Overpotential (overvoltage)	Extra voltage needed (beyond E°) to discharge **gases** — especially O_2 — at an electrode. This is why Cl_2 is often produced instead of O_2 despite E° predictions.
Nature of the electrode	Inert electrodes (Pt, graphite) are not consumed. **Active** electrodes (e.g. Cu) can themselves be oxidised at the anode instead of the solution.

WORKED EXAMPLE — ELECTROLYSIS OF AQUEOUS $CuSO_4$ WITH INERT ELECTRODES

Cathode: Cu^{2+} (E° = +0.34) vs H_2O (E° = -0.83) Cu^{2+} wins **Cu(s) deposited**

Anode: SO_4^{2-} is very hard to oxidise, so water is oxidised **O_2(g)** + H^+ Result: copper plates onto the cathode; oxygen bubbles at the anode; the solution becomes acidic.

11 Faraday's Laws — Electrolysis Calculations

The **amount** of substance produced at an electrode is directly proportional to the quantity of electric charge passed. This lets us calculate masses, moles and volumes.

ESSENTIAL FORMULAE

Formula	Meaning of terms
$Q = I \times t$	Q = charge (coulombs, C); I = current (amperes, A); t = time (seconds, s)
$n(e^-) = Q \div F$	F = Faraday constant = **96 500 C mol^{-1}**
use the half-equation mole ratio	relate mol e^- to mol of product
$m = n \times M$ or $V = n \times V_m$	convert moles to mass or gas volume

STANDARD 5-STEP METHOD

1. Calculate charge: **Q = It** (convert time to seconds!).
2. Calculate moles of electrons: **$n(e^-) = Q / 96\ 500$**.
3. Write the electrode half-equation and read the **electron : product ratio**.
4. Calculate moles of product using that ratio.
5. Convert to the required quantity (mass $m = nM$, or gas volume $V = nV_m$).

WORKED EXAMPLE — SILVER ELECTROPLATING

A current of **2.00 A** flows for **30.0 minutes** through $AgNO_3$. What mass of silver is deposited?

Step 1: $Q = It = 2.00 \times (30.0 \times 60) = 2.00 \times 1800 = 3600$ C

Step 2: $n(e^-) = 3600 \div 96\ 500 = 0.03731$ mol

Step 3: $Ag^+ + e^-\quad Ag$ (ratio 1 e^- : 1 Ag)

Step 4: $n(Ag) = 0.03731$ mol

Step 5: $m = n \times M = 0.03731 \times 107.9 =$ **4.03 g of silver**

WORKED EXAMPLE — GAS VOLUME AT THE CATHODE

Same charge (3600 C) in the electrolysis of water. Volume of H_2 at STP ($V_m = 24.79$ L mol^{-1} at 25 °C, 100 kPa)? $2H_2O + 2e^-\quad H_2 + 2OH^-$ (ratio 2 e^- : 1 H_2)

$n(H_2) = 0.03731 \div 2 = 0.01865$ mol

$V = 0.01865 \times 24.79 =$ **0.463 L (463 mL)**

CALCULATION TRAPS

- Always convert **minutes/hours** **seconds** before using Q = It.
- Watch the **electron ratio**: Cu^{2+} needs 2 e^-, Al^{3+} needs 3 e^- — this changes the moles of product.
- If two cells are in **series**, the **same charge** passes through both.

12 Industrial Applications

Electroplating

Coating an object with a thin metal layer for protection or appearance. The **object is the cathode**; the **pure plating metal is the anode**; the electrolyte contains ions of the plating metal.

Cathode (object): $M^{n+} + n\,e^-$ M(s) Anode (pure metal): M(s) $M^{n+} + n\,e^-$

Electrorefining of copper

Impure copper is the **anode** and dissolves; pure copper deposits on the **cathode**. Valuable impurities (Ag, Au) collect below the anode as "anode sludge"; more reactive metals stay in solution.

Extraction of reactive metals (e.g. aluminium — Hall–Héroult process)

Very reactive metals (Al, Na, Mg, K) cannot be extracted by carbon reduction, so they are obtained by **electrolysis of the molten compound**. Aluminium oxide (Al_2O_3) is dissolved in molten cryolite to lower the melting point.

Cathode: $Al^{3+} + 3e^-$ Al(l) Anode: $2O^{2-}$ $O_2 + 4e^-$ (carbon anodes burn away as CO_2)

13 Corrosion of Iron & Its Prevention

Corrosion is the unwanted oxidation of a metal by its environment. Rusting of iron is an electrochemical process requiring **both water and oxygen**; it is accelerated by dissolved salts (electrolytes) and acids.

THE ELECTROCHEMISTRY OF RUSTING

Anodic region (oxidation): Fe(s) $Fe^{2+}(aq) + 2e^-$

Cathodic region (reduction): $O_2(g) + 2H_2O(l) + 4e^-$ $4OH^-(aq)$ Fe^{2+} is then further oxidised by O_2 and precipitates as hydrated iron(III) oxide, $Fe_2O_3{\cdot}xH_2O$ — **rust**.

Prevention methods

Method	How it works
Barrier coatings (paint, oil, plastic)	Physically excludes water and oxygen. Fails if the coating is scratched.
Galvanising (zinc coating)	Dual action: barrier + **sacrificial protection**. Even if scratched, Zn (more reactive, more negative E°) oxidises in preference to Fe.
Sacrificial anode (block of Zn or Mg)	A more reactive metal is attached; it is oxidised instead of the iron. Used on ship hulls and pipelines. Anode is "sacrificed" and replaced periodically.
Cathodic protection (impressed current)	An external power source forces the iron to be the **cathode**, so it cannot be oxidised.

Alloying (stainless steel)	Adding Cr/Ni forms a protective, self-repairing oxide layer.

WHY SACRIFICIAL PROTECTION WORKS

A metal with a **more negative E°** than iron (e.g. Zn -0.76 V, Mg -2.37 V vs Fe -0.44 V) is a stronger reducing agent, so it is preferentially oxidised, protecting the iron even where the coating is broken.

14 Cells in the Real World

Primary (non-rechargeable)

Reaction is not reversible; discarded when flat. E.g. dry cell (Zn/MnO_2), alkaline cell.

Secondary (rechargeable)

Reaction is reversed by an external supply (recharging is electrolysis). E.g. lead–acid battery, lithium-ion cell.

Fuel cells

Reactants supplied continuously; e.g. H_2/O_2 fuel cell producing water. High efficiency, low pollution.

HYDROGEN–OXYGEN FUEL CELL (ACIDIC)

Anode: $2H_2 \quad 4H^+ + 4e^-$ Cathode: $O_2 + 4H^+ + 4e^- \quad 2H_2O$

Overall: $2H_2 + O_2 \quad 2H_2O$

15 Galvanic vs Electrolytic — Master Comparison

	Galvanic (Voltaic) Cell	Electrolytic Cell
Purpose	Generates electricity	Uses electricity to drive a reaction
Energy conversion	Chemical electrical	Electrical chemical
ΔG / $E°_{cell}$	$\Delta G < 0$, $E°_{cell} > 0$ (spontan-eous)	$\Delta G > 0$, $E°_{cell} < 0$ (non-spontaneous)
Power source	None (is the source)	External battery / power supply required
Oxidation occurs at	Anode	Anode
Reduction occurs at	Cathode	Cathode
Anode sign	Negative (-)	Positive (+)
Cathode sign	Positive (+)	Negative (-)
Electron flow (wire)	Anode cathode	Anode cathode
Examples	Batteries, fuel cells	Electroplating, electrorefining, Al/NaCl extraction, rechar-ging

16 Exam Skills, Common Mistakes & Glossary

TOP EXAM TRAPS TO AVOID

- Confusing the **agent** with the **process**: the oxidising agent is *reduced*.
- Changing E° when you multiply a half-equation — **never** do this (E° is intensive).
- Forgetting the salt bridge's job = **maintain electrical neutrality / complete the circuit** (not "carry electrons").
- Mixing up cell polarity — remember only the **+/-** flips between cell types; oxidation is always at the anode.
- Not converting **time to seconds** in $Q = It$.
- Ignoring **water** as a competing species in aqueous electrolysis.
- Not **checking charge balance** when balancing/combining half-equations.
- Assuming a positive $E°_{cell}$ means the reaction is fast — it only means it is **feasible**.

HOW TO ANSWER "PREDICT AND EXPLAIN A REACTION" QUESTIONS

1. Identify all possible half-equations from the data sheet.
2. Pick the strongest oxidant (highest E°) and strongest reductant (lowest E°).
3. Reverse the oxidation half-equation and add so electrons cancel.
4. Calculate $E°_{cell} = E°_{cathode} - E°_{anode}$; if positive, the reaction occurs.
5. State observations (colour change, gas, deposit, mass change).

Glossary of key terms

Term	Definition
Redox reaction	Reaction involving transfer of electrons (simultaneous oxidation and reduction).
Oxidation	Loss of electrons; increase in oxidation number.
Reduction	Gain of electrons; decrease in oxidation number.
Oxidising agent	Electron acceptor; is itself reduced.
Reducing agent	Electron donor; is itself oxidised.
Oxidation number	Charge an atom would have if all bonds were ionic; tracks electron transfer.
Half-equation	Equation showing only oxidation or only reduction, including electrons.
Conjugate redox pair	Oxidised and reduced forms of a species linked by e^- transfer.
Anode	Electrode where oxidation occurs.
Cathode	Electrode where reduction occurs.
Salt bridge	Maintains electrical neutrality and completes the circuit in a galvanic cell.
Standard electrode potential (E°)	Reduction potential of a half-cell vs the SHE under standard conditions.
SHE	Standard Hydrogen Electrode; reference half-cell defined as 0.00 V.
Electrochemical series	List of half-reactions ordered by E° (oxidising strength).
Electrolysis	Using electrical energy to drive a non-spontaneous redox reaction.
Overpotential	Extra voltage (above E°) needed to discharge a species, esp. gases.
Faraday constant (F)	Charge of one mole of electrons = 96 500 C mol^{-1}.
Disproportionation	One element is simultaneously oxidised and reduced.

A Appendix — Standard Reduction Potentials (25 °C)

All half-equations are written as **reductions**. Strongest oxidising agents are at the top (most positive E°); strongest reducing agents (reduced forms) are at the bottom (most negative E°). Values are indicative — always use the exact values on your exam data sheet.

Ordered from strongest oxidant (top) to strongest reductant (bottom)

Reduction half-equation	E° (V)	Note
$F_2(g) + 2e^- \quad 2F^-$	+2.87	strongest oxidant
$H_2O_2 + 2H^+ + 2e^- \quad 2H_2O$	+1.77	
$MnO_4^- + 8H^+ + 5e^- \quad Mn^{2+} + 4H_2O$	+1.51	acidified permanganate
$Cl_2(g) + 2e^- \quad 2Cl^-$	+1.36	
$Cr_2O_7^{2-} + 14H^+ + 6e^- \quad 2Cr^{3+} + 7H_2O$	+1.33	acidified dichromate
$O_2(g) + 4H^+ + 4e^- \quad 2H_2O$	+1.23	
$Br_2(l) + 2e^- \quad 2Br^-$	+1.09	
$Ag^+ + e^- \quad Ag(s)$	+0.80	
$Fe^{3+} + e^- \quad Fe^{2+}$	+0.77	
$O_2(g) + 2H_2O + 4e^- \quad 4OH^-$	+0.40	
$Cu^{2+} + 2e^- \quad Cu(s)$	+0.34	
$Sn^{4+} + 2e^- \quad Sn^{2+}$	+0.15	
$2H^+ + 2e^- \quad H_2(g)$	**0.00**	**SHE reference**
$Pb^{2+} + 2e^- \quad Pb(s)$	-0.13	
$Sn^{2+} + 2e^- \quad Sn(s)$	-0.14	
$Ni^{2+} + 2e^- \quad Ni(s)$	-0.25	
$Fe^{2+} + 2e^- \quad Fe(s)$	-0.44	
$Zn^{2+} + 2e^- \quad Zn(s)$	-0.76	used in galvanising
$2H_2O + 2e^- \quad H_2(g) + 2OH^-$	-0.83	water reduction
$Al^{3+} + 3e^- \quad Al(s)$	-1.66	
$Mg^{2+} + 2e^- \quad Mg(s)$	-2.37	
$Na^+ + e^- \quad Na(s)$	-2.71	
$Ca^{2+} + 2e^- \quad Ca(s)$	-2.87	
$K^+ + e^- \quad K(s)$	-2.93	
$Li^+ + e^- \quad Li(s)$	-3.04	strongest reductant

USEFUL CONSTANTS & DATA

- Faraday constant $F = 96\,500 \text{ C mol}^{-1}$
- Charge on one electron = 1.602×10^{-19} C
- Avogadro constant $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$
- Molar gas volume $V_m = 24.79 \text{ L mol}^{-1}$ (25 °C, 100 kPa) · 22.71 L mol^{-1} (0 °C, 100 kPa)
- $Q = It$ · $n(e^-) = Q/F$ · $E°_{cell} = E°_{cathode} - E°_{anode}$ · $\Delta G° = -nFE°_{cell}$

End of notes — Redox & Electrochemistry · ATAR Year 12 Chemistry. Review the data sheet and worked examples regularly, and practise balancing half-equations under timed conditions.